OWASP | TOP 10 LLM APPLICATIONS & GENERATIVE AI

genai.owasp.org

LLM and Gen AI Security Solution Landscape Guide

CHEAT SHEET - SERIES

Q1, 2025

https://genai.owasp.org/ai-security-solutions-landscape/

OWASP | TOP 10 LLM APPLICATIONS & GENERATIVE AI

CHEAT SHEET

LLM & GenAI Security Landscape - 2025, Q1

https://genai.owasp.org/ai-security-solutions-landscape/

Open Source

Operate

aqua, securiti AI Security & Governance, Llama Guard Meta, ZenGuard AI, NRI SECURE AI Blue Team, cisco AI Firewall®, LLM GUARD, BLUETEAM AI, aim security, PRISMA, TROJ.AI, paloalto NETWORKS

Deploy

TROJ.AI, cisco AI Firewall®, BLUETEAM AI, Meta, paloalto NETWORKS, securiti AI Security & Governance, Operant, PRISMA

Release

securiti AI Security & Governance, cisco AI Validation, CycloneDX, PRISMA, LEGIT SECURITY AI-SPM, paloalto NETWORKS, aqua

Test & Evaluate

pillar SECURITY, cisco AI Validation, Prompt Fuzzer, promptfoo, aqua, Enkrypt AI, Giskard, Synack, LangCheck, MODELSCAN, securiti AI Security & Governance, CYBERSECEVAL Meta, ZenGuard AI, LLM vulnerability scanner, HARMBENCH, Cobalt, PROTECT AI, WATCHTOWER BY AISHIELD, PRISMA, MINDGARD, TROJ.AI, Citadel AI, AIShield Powered by Bosch, Cymetrics

Develop & Experiment

Meta, Mend AI, TRAIL OF BITS, aqua, cisco AI Validation, snyk, securiti AI Security & Governance, pangea, Operant, PRIVACY Raven, Cloaked AI, TROJ.AI, PRISMA

Augment, Fine Tune Data

Cloaked AI, securiti AI Security & Governance, UNSTRUCTURED.IO, BLUETEAM AI, PRISMA

Scope & Plan

MITRE ATLAS, BLUETEAM AI, Seezo, paloalto NETWORKS, pillar SECURITY, securiti, PRISMA, STRIDE GPT

Monitor

aqua, cisco AI Validation, aim security, BLUETEAM AI, AIShield Powered by Bosch, HIDDENLAYER, LAKERA, Meta Prompt Guard, Operant, paloalto NETWORKS, PROTECT AI, securiti, splx AI, TROJ.AI, ZenGuard AI

Govern

aim security, cisco AI Validation, BLUETEAM AI, Lasso SECURITY, paloalto NETWORKS, PRISMA, Prompt:, securiti, Tumeryk AI GUARD, unbound security, AI VERIFY FOUNDATION

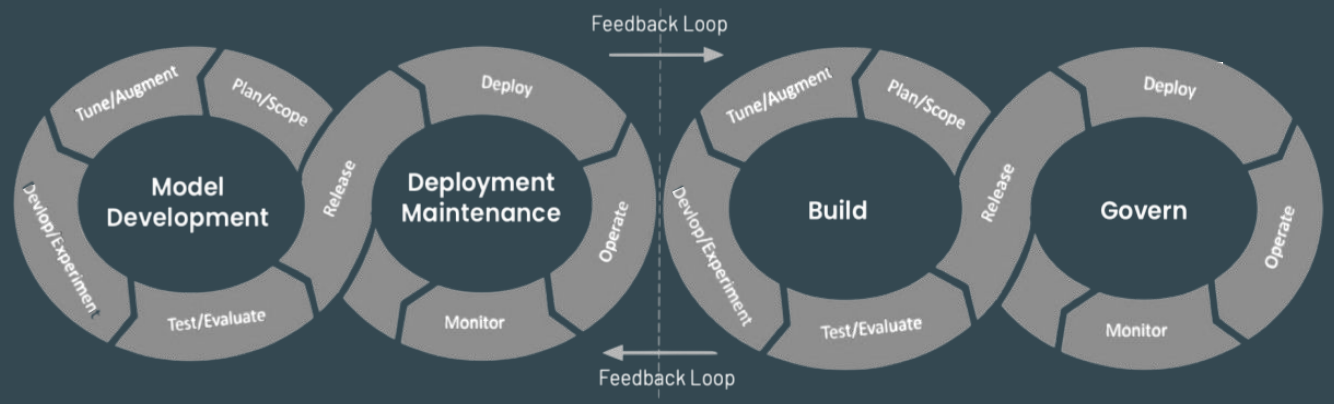
Source; OWASP Gen AI Security Solutions Landscape Guide 2025. Q1

OWASP | TOP 10 LLM APPLICATIONS & GENERATIVE AI

CHEAT SHEET

LLM and Gen AI App SecOps Framework

https://genai.owasp.org/ai-security-solutions-landscape/



The OWASP LLMSecOps Framework was captured to help better align LLMOps processes and the security roles and dependencies for each stage. While LLMOps and MLOps are rooted in the same foundational principles of lifecycle management, they can diverge significantly in their focus and requirements, as one is focused primarily on model development, while the other extends the model and is application to various Gen AI application patterns.

Plan & Scope

- Access Control and Authentication Planning
- Compliance and Regulatory Assessment
- Data Privacy and Protection Strategy
- Early Identification of Sensitive Data
- Third-Party Risk Assessment (Model, Provider, etc.)
- Threat Modeling
-

Augment & Fine Tune Data

- Data Source Validation
- Secure Data Handling
- Secure Output Handling
- Adversarial Robustness Testing
- Model Integrity Validation (ex: serialization scanning for malware)
- Vulnerability Assessment

Dev & Experiment

- Access, Authentication, and Authorization (MFA)
- Experiment Tracking
- LLM & App Vuln Scanning
- Model and Application Interaction Security
- SAST/DAST
- Secure Coding Practices
- Secure Library/Code Repository
- Software Comp Analysis

Test & Evaluation

- Adversarial Testing
- Application Security Orchestration and Correlation
- Bias and Fairness Testing
- Final Security Audit
- Incident Simulation, Response Testing
- LLM Benchmarking
- Penetration Testing
- IAST
- Vulnerability Scanning

Release

- AI/ML Bill of Materials (BOM)
- Digital Model\Dataset Signing
- Model Security Posture Evaluation
- Secure CI/CD pipeline
- Secure Supply Chain Verification
- Static and Dynamic Code Analysis
- User Access Control Validation
- Model Serialization Defenses

Deploy

- Compliance Verification
- Deployment Validation
- Digital Model\Dataset Verification
- Encryption, Secrets management
- Multi-factor Authentication
- Network Security Validation
- Secure API Access
- Secure Configuration
- User and Data Privacy Protections

Operate

- Adversarial Attack Protection
- Automated Vuln Scanning
- Data Integrity and Encryption
- LLM Guardrails
- LLM Incident Detection and Response
- Patch Management
- Privacy, Data Leakage Protection
- Prompt Security
- Runtime Self-Protection
- Secure Output Handling

Monitor

- Adversarial Input Detection
- Model Behavior Analysis
- AI/LLM Secure Posture Management
- Patch and Update Alerts
- Regulatory Compliance Tracking
- Security Alerting
- Security Metrics Collection
- User Activity Monitoring
- Observability
- Data Privacy and Protection
- Ethical Compliance

Govern

- Bias and Fairness Oversight
- Compliance Management
- Data Security Posture Management
- Incident Governance
- Risk Assessment and Management
- User/Machine Access audits

Source; OWASP Gen AI Security Solutions Landscape Guide 2025. Q1

OWASP | TOP 10 LLM APPLICATIONS & GENERATIVE AI

CHEAT SHEET SERIES

Project Sponsors

Gold: HIDDENLAYER, paloalto NETWORKS, snyk, Mend.io

Silver: Cobalt, pangea, Prompt, Synack, Lasso SECURITY, securiti, PROMPTARMOR

Project Supporters

Accenture	BeDisruptive	Comcast	GitHub	Lakera	NuBinary	Securiti	VE3
AddValueMachine Inc	Bit79	Complex Technologies	Google	Lasso Security	Palo Alto Networks	Securiti AI	WhyLabs
Aeye Security Lab Inc.	Blue Yonder	Credal.ai	GuidePoint Security	Layerup	Palosade	See-Docs & Thenavigo	Yahoo
AI informatics GmbH	BroadBand Security, Inc.	Databook	HackerOne	Legato	Praetorian	ServiceTitan	
AI Village	BuddoBot	DistributedApps.ai	HADESS	Linkfire	Preamble	SHI	
aigos	Bugcrowd	DreadNode	IBM	LLM Guard	Precize	Smiling Prophet	
Aon	Cadea	DSI	iFood	LOGIC PLUS	Prompt Security	Snyk	
Aqua Security	Check Point	EPAM	IriusRisk	MaibornWolff	PromptArmor	Sourcetoad	
Astra Security	Cisco	Exabeam	IronCore Labs	Mend.io	Pynt	Sprinklr	
AVID	Cloud Security Podcast	EY Italy	IT University Copenhagen	Microsoft	Quiq	stackArmor	
AWARE7 GmbH	Cloudflare	F5	Kainos	Modus Create	Red Hat	Tietoevry	
AWS	Cloudsec.ai	FedEx	KLAVAN	Nexus	RHITE	Trellix	
BBVA	Coalfire	Forescout	Klavan Security Group	Nightfall AI	SAFE Security	Trustwave SpiderLabs	
Bearer	Cobalt	GE HealthCare	KPMG Germany FS	Nordic Venture Family	Salesforce	U Washington	
BeDisruptive	Cohere	Giskard	Kudelski Security	Normalyze	SAP	University of Illinois	

View the Latest Project Sponsors and Supporters on the Web: https://genai.owasp.org/supporters/

https://genai.owasp.org

scott.clinton@owasp.org